

EDUCATION

- Indian Institute of Technology
 - AVB Public School

B.S + M.S. in Mathematics and Computing (Integrated)

CBSE Senior School Certificate Examination

Kharagpur, India

Delhi, India

July 2015 - June 2020

2015

PROFESSIONAL EXPERIENCE

- Microsoft
 - Microsoft
 - Samsung Research Institute
 - Prakshep
 - Farebond

Software Engineer 2

Software Engineer

Research Intern

Data Science Intern

Data Science Intern

Hyderabad, India

Hyderabad, India

Bengaluru, India

Bengaluru, India

Bengaluru, India

Mar 2023 - Present

Jul 2020 - Mar 2023

May 2019 - July 2019

May 2018 - June 2018

May 2017 - Aug 2017
- Designed & implemented config-based component architecture, enabling developers for faster iterations and minimal setup
 - Led & mentored engineers to migrate the News Section to new service, redesigning the UI for enhanced customer experience
 - Core contributor to the Viva Connections app, driving development and scalability to support over **9M+ MAU**
 - Architected critical features with React Native and TypeScript, directly improving the experience for **1M+ users**
 - Resolved high-priority customer issues within SLA timelines, increasing customer satisfaction
 - Championed the Day of Learning initiative, fostering a culture of continuous technical growth within the team

- Implemented data partitioning and scoping paradigm across application for **5+ scenarios** with shared JS runtime
 - Developed robust link interception module, directing **2M+ users** to correct in-app experiences with **99.9%** reliability
 - Collaborated cross-functionally with designers and product managers across geographies to deliver features
 - Created in-app developer tools to enhance developer experience, saving up to **60+ man-hours monthly**

- Developed Deep Learning solution to eliminate calibration process for wearable controllers using TensorFlow and Keras
 - Engineered Recurrent Neural Network (RNN) model to predict calibration parameters, achieving **75% reduction in error**
 - Formulated specialized loss function to accurately quantify angular discrepancies in IMU sensor arrays
 - Implemented advanced data augmentation techniques to represent quaternion space fully, increasing model robustness

- Analyzed satellite imagery using Python to study moisture trends in agricultural farms, providing actionable insights
 - Processed Sentinel-2 satellite images and applied image processing techniques to classify water bodies, agricultural lands
 - Designed statistical model based on Z-scores of spectral indices to identify optimal and outlier plots
 - Developed RESTful APIs using Flask and Python to compute central tendency measures and implement cloud detection

- Contributed to option pricing model project focused on pricing fare locks for flights, enhancing booking flexibility
 - Designed tier-based pricing model categorizing flights based on fare volatility, assigning optimized prices
 - Developed automated reporting system using R and R Markdown, leading to time savings of **60+ man-hours monthly**
 - Automated generation of interactive visualizations using R for analyzing data to monitor anomalies and trends

TECHNICAL SKILLS

- Languages: JavaScript, TypeScript, Python,

- C++, Swift, C#
 - Frameworks: React, React Native, .NET

- Tools: Git, CI/CD Pipelines, KQL, TensorFlow, Keras

- Databases: SQL, NoSQL

PROJECTS

- MacOS Themed Playground: [WIP]Interactive Desktop Environment
 - Portfolio Website: Personal Website in React
 - EthPhoto: Decentralized Web Application

June 2025

April 2025

Mar 2017
- Built pixel-perfect macOS UI replica using React 18, TypeScript, and styled-components and custom animations
 - Developed Go backend with RESTful APIs serving JSON configurations and static assets using http.FileServer middleware
 - Implemented React Context API for global state management handling window operations and application lifecycle
 - Built responsive components with TypeScript interfaces: draggable windows, auto-hiding menubar, and dock animations

- Implemented data-driven component architecture with clear separation between presentation and content
 - Engineered custom integrations with external data sources (Medium RSS) with fallback mechanisms
 - Implemented a GraphQL data layer to query & transform Markdown content, creating structured content management system
 - Used AI Tools such as Copilot and Claude to quickly understand and implement features

- Developed decentralized application for geo-tagged photo sharing on Ethereum blockchain using Solidity
 - Integrated with InterPlanetary File System (IPFS) to store photos and metadata, saving hashes on blockchain
 - Contributed to development of responsive GUI using JavaScript, HTML, and CSS, enhancing user experience